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Math 3108 – Linear Algebra

Exam 1

Prof. Runnion

September 13, 2024

Directions:

- 1) All cell phones and other electronic noisemaking devices must be turned completely off (silence or on vibrate is *not* acceptable) and put away for the duration of the exam.
- 2) No calculators, books, or other materials are permitted.
- 3) You must write **darkly** and **legibly** – this exam will be scanned for electronic grading.
- 4) Show **ALL** your work! Correct answers which are not properly justified will not receive full credit.
- 5) Failure to follow directions specific to a problem will result in the loss of points.
- 6) Circle, box, or underline each final answer if necessary for clarity. Label all graphs.
- 7) Somewhere on this exam, it will be helpful to remember that $T(\mathbf{0}) = \mathbf{0}$ for every linear transformation T .
- 8) If you work a problem two different ways, clearly indicate which one you want us to grade, preferably by crossing out the one you do not want us to look at. If you use two methods, one of which is wrong, and neither method is crossed out, you will not receive full credit.
- 9) Answers must be exact (like $\sqrt{2}$), not approximate (like 1.414), unless a problem specifically indicates otherwise.
- 10) Simplify where appropriate. Quantities such as $\sqrt{9}$ and $\cos \pi$ should be calculated.
- 11) If A has a pivot in every column, then the transformation $T(\mathbf{x}) = A\mathbf{x}$ is one-to-one.
- 12) Extra space has been provided on all problems. There should be more than sufficient space to work all problems in the space indicated. Thus, **NO WORK APPEARING ON THE BACK OF PAGES WILL BE GRADED!**
- 13) All work must appear within this exam packet to be graded. **No scratch paper is allowed under any circumstances.**
- 14) This packet has six sheets of paper, including this cover page. Do NOT remove the staple or remove any sheet from this packet.
- 15) Once this exam begins, you will have 50 minutes to complete your solutions.

DO NOT OPEN THIS EXAM UNTIL TOLD TO DO SO

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Multiple Choice and True/False. Fill in the box corresponding to the correct answer.

1. (10 points) Consider the matrix $A = \begin{bmatrix} 1 & -4 & 0 & 2 \\ 0 & 2 & 3 & 1 \\ 0 & 0 & 6 & 2 \end{bmatrix}$.

i) A is in _____.

☒ Echelon Form ☐ Reduced Echelon Form ☐ Neither

ii) The equation $A\mathbf{x} = \mathbf{0}$ has a nontrivial solution.

☒ True ☐ False

iii) The columns of A span _____.

☐ $\mathbb{R}^2$ ☒ $\mathbb{R}^3$ ☐ $\mathbb{R}^4$ ☐ None of these

iv) The transformation $T(\mathbf{x}) = A\mathbf{x}$ is _____ and _____.

☐ One-to-One ☒ Not One-to-One

☒ Onto ☐ Not Onto

2. (10 points) Consider the matrix $B = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & -4 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$.

i) B is in _____.

☐ Echelon Form ☒ Reduced Echelon Form ☐ Neither

ii) The equation $B\mathbf{x} = \mathbf{0}$ has a nontrivial solution.

☒ True ☐ False

iii) The columns of B span _____.

☒ $\mathbb{R}^2$ ☐ $\mathbb{R}^3$ ☐ $\mathbb{R}^4$ ☐ None of these

iv) The transformation $T(\mathbf{x}) = B\mathbf{x}$ is _____ and _____.

☐ One-to-One ☒ Not One-to-One

☐ Onto ☒ Not Onto

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3. (2 points) If the set $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\}$ is a set of linearly independent vectors in $\mathbb{R}^7$, then the set $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is also linearly independent.

☒ a) Always True ☐ b) Sometimes True ☐ c) Never True

4. (2 points) If $\mathbf{u}$ and $\mathbf{v}$ are vectors in $\mathbb{R}^5$, then the vector $3\mathbf{u}$ is in $\text{Span}\{\mathbf{u}, \mathbf{v}\}$.

☒ a) Always True ☐ b) Sometimes True ☐ c) Never True

Open Answer. Show ALL your work! Place your final answer in the box provided, if applicable.

5. (5 points each) Let $A = \begin{bmatrix} 1 & -3 \\ -2 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 0 & -5 \\ 1 & -3 & 4 \end{bmatrix}$, and $\mathbf{x} = \begin{bmatrix} 5 \\ 3 \end{bmatrix}$. For each of the following quantities, compute the quantity or, if it does not exist, write "DNE" inside the matrix brackets and clearly explain why it does not exist.

a) $AB = \begin{bmatrix} -1 & 9 & -17 \\ 0 & -12 & 26 \end{bmatrix}$ 2×3

b) $BA = \begin{bmatrix} \text{DNE} \end{bmatrix}$ The max matrix B is 2×3 , the nxp matrix is 2×2 , n has to equal n and $3 \neq 2$, so the quantity DNE.

c) $\mathbf{x}^T A^T = \begin{bmatrix} -12 & 14 \end{bmatrix}$ $5 \times -3 + 3 \times 1$ $5 \times 4 + 3 \times -2$

$\begin{bmatrix} 5 & 3 \end{bmatrix} \begin{bmatrix} -3 & 4 \\ 1 & -2 \end{bmatrix}$
 $1 \times 2 \quad 2 \times 2 = 1 \times 2$

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6. a) (16 points) Solve the matrix equation $A\mathbf{x} = \mathbf{b}$ using augmented matrices, where

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ -1 & 4 & 3 \end{bmatrix} \text{ and } \mathbf{b} = \begin{bmatrix} 2 \\ 7 \\ -8 \end{bmatrix}.$$

Show all your work! State your solution in parametric vector form.

$$\begin{aligned} &\left[\begin{array}{ccc|c} 1 & 2 & 3 & 2 \\ 2 & 1 & 3 & 7 \\ -1 & 4 & 3 & -8 \end{array} \right] \xrightarrow{\substack{-2R_1 + R_2 \rightarrow R_2 \\ R_1 + R_3 \rightarrow R_3}} \left[\begin{array}{ccc|c} 1 & 2 & 3 & 2 \\ 0 & -3 & -3 & 3 \\ 0 & 6 & 6 & -6 \end{array} \right] \xrightarrow{2R_2 + R_3 \rightarrow R_3} \left[\begin{array}{ccc|c} 1 & 2 & 3 & 2 \\ 0 & -3 & -3 & 3 \\ 0 & 0 & 0 & 0 \end{array} \right] \\ &\xrightarrow{-\frac{1}{3}R_2} \left[\begin{array}{ccc|c} 1 & 2 & 3 & 2 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{array} \right] \xrightarrow{-2R_2 + R_1 \rightarrow R_1} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 4 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{array} \right] \end{aligned}$$

$x_1 + x_3 = 4$ $x_1 = -x_3 + 4$
 $x_2 + x_3 = -1$ $x_2 = -x_3 - 1$
 x_3 is free

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -x_3 + 4 \\ -x_3 - 1 \\ x_3 \end{bmatrix}$$

Solution: $\mathbf{x} = \begin{bmatrix} 4 \\ -1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}$

- b) (6 points) Without performing any additional row operations, state the solution set for the homogeneous equation $A\mathbf{x} = \mathbf{0}$ where A is the matrix given in part (a) above.

Solution: $\mathbf{x} = x_3 \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}$

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7. (11 points) Let $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a linear transformation where $T(\mathbf{u}) = \begin{bmatrix} -1 \\ 4 \end{bmatrix}$ and

$$T(\mathbf{v}) = \begin{bmatrix} 2 \\ 3 \end{bmatrix}. \text{ Use the fact that } T \text{ is linear to find } T(\mathbf{u} - 2\mathbf{v}).$$

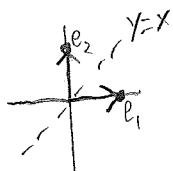
$$T(\mathbf{u} - 2\mathbf{v}) = T(\mathbf{u}) - 2T(\mathbf{v}) = \begin{bmatrix} -1 \\ 4 \end{bmatrix} - 2 \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} -1 \\ 4 \end{bmatrix} - \begin{bmatrix} 4 \\ 6 \end{bmatrix} = \begin{bmatrix} -5 \\ -2 \end{bmatrix}$$

Solution: $T(\mathbf{u} - 2\mathbf{v}) = \begin{bmatrix} -5 \\ -2 \end{bmatrix}$

8. (12 points) Find the standard matrix A of the linear transformation $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ which first rotates points counterclockwise about the origin by an angle of $\varphi = \frac{\pi}{2}$ and then reflects points about the line $y = x$.

(Hint: Think about what this transformation does to $\mathbf{e}_1$ and $\mathbf{e}_2$.)

$$\begin{matrix} \mathbf{e}_1 & \mathbf{e}_2 \\ \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \end{matrix} \xrightarrow{\varphi = \frac{\pi}{2}} \begin{matrix} \mathbf{e}_1 & \mathbf{e}_2 \\ \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \end{matrix} \xrightarrow{y=x} \begin{matrix} \mathbf{e}_1 & \mathbf{e}_2 \\ \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \end{matrix}$$



Solution: $A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$

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9. (8 points) The following *augmented* matrix in echelon form represents a system of linear equations. If the system is consistent, find the general solution in parametric vector form and place it in the box. Otherwise, leave the box blank, state that the system is inconsistent, and indicate why you came to that conclusion.

$$\begin{bmatrix} 1 & 0 & 4 & 6 \\ 0 & 1 & 0 & 4 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

The system is inconsistent because there is a pivot in the rightmost column.

Solution:

$$\mathbf{x} = \begin{bmatrix} \\ \\ \end{bmatrix} + \begin{bmatrix} \\ \\ \end{bmatrix}$$

10. (8 points) Show that the transformation T defined by $T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} 2x_1 - 3x_2 \\ x_1 + 4 \\ 5x_2 \end{bmatrix}$ is not linear.

$$T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} 2 & -3 & 0 \\ 1 & 0 & 4 \\ 0 & 5 & 0 \end{bmatrix} \xrightarrow{\text{Swap } R_1 + R_2} \begin{bmatrix} 1 & 0 & 4 \\ 2 & -3 & 0 \\ 0 & 5 & 0 \end{bmatrix} \xrightarrow{-2R_1 + R_2 \rightarrow R_2} \begin{bmatrix} 1 & 0 & 4 \\ 0 & -3 & -8 \\ 0 & 5 & 0 \end{bmatrix}$$

$$\xrightarrow{\frac{5}{3}R_2 + R_3 \rightarrow R_3} \begin{bmatrix} 1 & 0 & 4 \\ 0 & -3 & -8 \\ 0 & 0 & -\frac{40}{3} \end{bmatrix} \xrightarrow{-\frac{1}{3}R_2} \begin{bmatrix} 1 & 0 & 4 \\ 0 & 1 & \frac{8}{3} \\ 0 & 0 & -\frac{40}{3} \end{bmatrix} \quad \begin{array}{l} x_1 = 4 \\ x_2 = \frac{8}{3} \\ 0 \neq -\frac{40}{3} \end{array}$$